Software Project Management

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ABSTACT
Software project management is the art of science of planning and leading software projects. It is sub-discipline of project management in which software project is planned, monitor and proscribed. The project objectives define the goal status at the end of project. Objectives are detailed, computable, achievable, and relevant and time constraint. Specific means effective objective. Unclear aspirations such as "to improve customer relations" are unacceptable. Objective is supposed to be defined in such way that it is obvious to all whether the project has been successful or not. Assessable tells that how much the scheme is successful. For example, 'to reduce customer complaints' would be more satisfactory as an aim than 'to improve client relations'. Achievable must be within the power of the personality or group to achieve the objective. Conformity with all stakeholders what the goal should be. Is there a realistic path to accomplishment the objective must be relevant to the true purpose of the project. There should be a defined point in time by which the aim should have been achieve.
Project management is ―the application of facts, skills, tools and techniques to project performance to meet the project requirements.‖ the effectiveness of project organization is critical in assuring the success of any substantial activity. Areas of responsibility for the person handling the project contain planning, control and implementation. A project should be initiate with a feasibility study, where a clear definition of the goals and ultimate benefits need to be unwavering. Senior manager support for projects is important so as to ensure authority and direction throughout the project's progress and, also to guarantee that the goals of the organization are effectively achieved in this process. Acquaintance, skills, goals and personalities are the factors that need to be measured within project management. The project manager and his/her team should together possess the necessary and requisite interpersonal and technological skills to facilitate control over the various activities within the project. The stages of performance must be articulated at the project planning phase. Disaggregating the stages at its early point assists in the successful expansion of the project by providing a number of milestones that need to be accomplished for achievement. In addition to planning, the control of the evolving project is also a prerequisite for its success. Control requires plenty monitoring and feedback mechanisms by which senior management and project managers can measure up to progress against initial projections at each stage of the project. Monitoring and criticism also enable the project manager to anticipate problems and therefore take pre-emptive and corrective measures for the advantage of the project. Projects normally involve the introduction of a new organization of some kind and, in almost all cases, new methods and ways of doing things. This impacts the work of others: the "user". User communication is an important factor in the success of projects and, indeed, the degree of user involvement can influence the extent of bear for the project or its implementation plan. A project manager is the one who is to responsibility for establishing a communication in between the project line-up and the user. Thus one of the most essential qualities of the project manager is that of being a good conversationalist, not just within the project team itself, but with the rest of the institute and outside world as well.
Keywords:- *People, Product, Process, Project, Capability maturity model, Team player, Software Team*

1 INTRODUCTION:-

1.1 SOFTWARE PROJECT MANAGEMENT:-

Software project management focuses on the 4 **P's: People, Product, Process and Project.**

fig 1.1 software project management

1.1.1 The People:-

The most important factor in the victory of software project "Companies that sensibly manage their asset in people will prosper in the long run" Tim & Tom. development of motivated and highly skilled software people has always been significant for software

organizations. The "people-factor" is so important that SEI has developed People Management Capability Maturity Model (PM-CMM).

1.1.1.1 MANAGEMENT SCALE

PM-CMM:-

"To increase the readiness of s/w organizations to commence increasingly complex applications by helping to attract, grow, inspire, deploy, and retain the talent needed to improve their software progress capability" .In simple words to enhance the peoples capabilities through personnel development. Organizations that realize high levels of maturity in PM-CMM have a higher likelihood of implementing effective software engineering practices. Key Practice Areas of PM-CMM is Recruiting, Selection, Performance Management and Training. The players, team leaders, software teams who participate in the software process and the manner in which they are organized to perform effective software engineering.

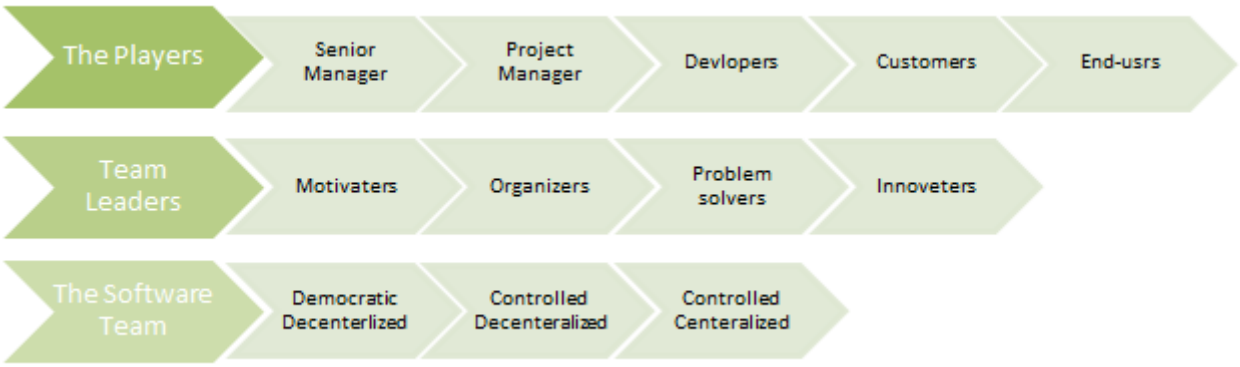

Fig 1.2 people

1.1.1.2 The People- (The Players)

They can be categorized into one of the subsequent Senior manager, project manager, practitioner, customer and end users.

Senior Managers define business issues that often have significant influence on production.

Project (technical) managers must plan, motivate, organize and control the practitioners who do software work. **Practitioners** deliver the scientific skills necessary to engineer a product or application .**Customers** specify the necessities for the software to be engineered. **End Users** interact with the software after it is released for production.

1.1.1.3 The People- (The Team Leaders)

Motivation encourage methodological people (by "push" or "pull") to produce required yield.

Organization apply and improve processes efficiently and one most significant thing is **Ideas or Innovation** which make people feel imaginative and be imaginative.

fig 1.3 team leaders

Characteristics of an effectual project managers is Problem Solving means to Diagnostic, Skill to solve ,Ability to design explanation etc. Managerial uniqueness means to Control the project.

Fig 1.4 team leaders

The Team Leaders - Characteristics of an valuable project managers is accomplishment means Reward Initiative ,Encourage Controlled danger taking authority and team building ,Influence the team, Read people"s mind and react according to their needs. Be controlled in stress situations.

1.1.1.4 The People- (The SoftwareTeam)
Organizations/configuration of teams:
Democratic decentralized, Controlled decentralized, Controlled centralized

Fig 1.5 software team
The Software Teams: **Democratic decentralized** which can have an explanation like No enduring leader ,Communication is horizontal Suitable for small projects requiring less than 5 to 6 engineers, research-oriented projects.

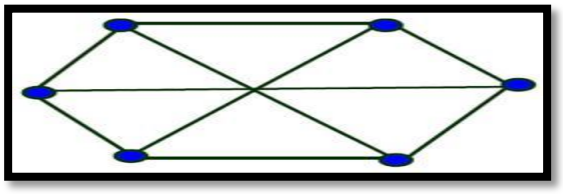

fig 1.6 Democratic decentralized
At different time, different members within the team supply technical leadership.
High morale and job fulfillment Due to autonomy, hence less employee turnover .
The Software Teams: **Controlled Centralized** is defined team leader, Problem solving, communication and managing by team leader and proclamation is vertical.

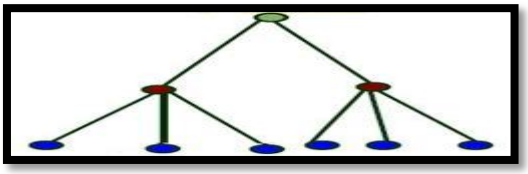

fig 1.7 Controlled Centralized
Team members may waste time in conflict about minor points due to deficiency of any influence in the team. The Software Teams: **Controlled Decentralized** draws upon the ideas from both former structures defined Leader Horizontal communication Problem solving is a group activity Suitable for large organization

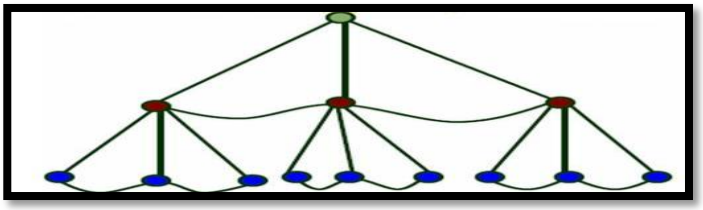

fig 1.8 Controlled Decentralized
The Software Teams ,complexity of task ,Size of resultant code (no. of lines) ,Time that team will stay together, Degree of modularization ,Required quality and dependability of the system being built and Rigidity of delivery date (schedule) ,Degree of communication.

1.1.1.5 The People - Agile Teams
Agile software development encourages consumer satisfaction and early incremental delivery of software with overall simplicity.
Agile teams are small, highly motivated teams.They adopt many distinctiveness of successful software project teams and avoid toxins that generate problems. They are self organizing and do not necessarily maintain a single team structure .Agile process models give significant sovereignty to agile teams.

fig 1.9 agile team
Planning is kept to bare minimum. The agile team is allowed to select its own approach (e.g., process, methods, tools). The agile team may have on a daily basis team meetings to synchronize and coordinate the day's work. With each passing day, this self business and collaboration move the team towards a accomplished software increment.

1.1.2 THE PRODUCT

Before a project should be designed, product objectives and scope should be established, alternative solutions should be considered, and technical and management constraints should be acknowledged. Software Scope is Problem Decomposition. Once the product objectives and scope are understood, alternative solutions are considered. The scope of product must be established and surrounded. Bounded scope means establishing quantitative data like no. of instantaneous users, max. allowable response time. etc. Constraints and boundaries and justifying factors described .

The problem that the product is addressing must be decomposed.

1.1.2.1 Software Scope

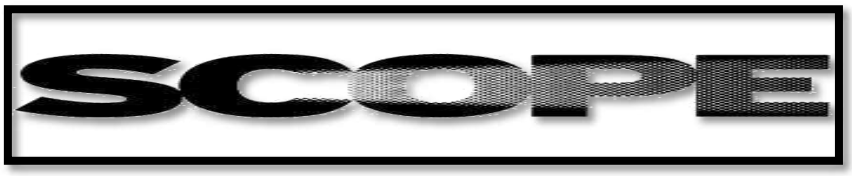

fig 1.10 software scope

Context :-Functional spot of the software product into a large system, manufactured goods or business context . Constraints involved:- Information Objectives :-What data objects are required as i/p or o/p .Function and Performance:- What function does the software structure execute on i/p to produce o/p. What level of performance is required Problem Decomposition

This activity is at core of requirements analysis Divide and conquer policy for complex troubles. A complex problem is partitioned into smaller problems that are more manageable.

Decomposition make planning easier. Decomposition in 2 major areas Functionality that must be delivered Process that will be used to deliver product.

1.1.3 THE PROCESS

Framework Activities ,Process Models ,Process putrefaction .Common Process Framework Activities characterize a software process and are applicable to all software projects Communication, Planning ,Modeling ,manufacture ,Deployment .

These are applied to software manufacturing work tasks (e.g., different product functions)

1.1.3.1 The Process Models

Linear sequential, Prototyping, RAD, Spiral

1.1.3.2 Project manager

They must decide about which model to use depending on ccustomers' who have requested the product , People who would work on project, Product characteristics ,Project situation, Project planning begins once model is selected. Process Decomposition is the way a process is decomposed depends on project complexity. Decomposition involves demarcation of work tasks involved in each process framework activity.

Example of disintegration for "communication" activity for a simple project:

Develop a list of clarification issues, Meet with customer to discuss clarification issues

Jointly develop statement of scope ,appraisal the declaration of scope with all concerned

Modify the statement of scope id required.

fig 1.11 software project

1.1.4 SOFTWARE PROJECTS

The software projects must be planned and forbidden effectively to avoid complexities.

The project managers and engineers must appreciate the critical success factors and develop a common sense approach for planning, monitor and controlling the project.

The software projects must be designed and controlled effectively to stay away from complexities.

The project managers and engineers must comprehend the critical success factors and develop a common sense come within reach for planning, monitoring and scheming the project.

1.1.4.1 Signs of Projects Risk

Software people don't understand buyer needs ,Product scope is poorly defined, Changes are managed poorly, The chosen technology changes ,Business needs change ,Deadlines are unrealistic ,Users are resistant ,funding is lost ,Team lacks skills ,Managers avoid best practices and so on.

1.1.4.2 How to shun problems?

Start on the right foot, Involves detailed thoughtful of project,setting realistic objectives & expectations ,Selecting the right team ,Facilitating the team, Maintain Momentum, Provide incentives, Reduce bureaucracy and give autonomy to team member but with supervision

Track Progress, Assess progress as work products are produced, Make smart decisions,When possible, use existing software components / COTS software ,Choose standard approaches and keep it simple and Avoid risks and allocate supplementary time than needed for complex/risky tasks. Conduct a postmortem analysis, Compare planned and actual schedule, Collect and analyze scheme metrics (standards), Get feedback from team and customers and create record of lessons learnt for each project.

2 Features of projects:

Projects are often approved out by a team of people who have been assembling for that specific purpose. The activities of this team may be synchronized by a project manager. Project teams may consist of persons from different background and different parts of the organisation. In some cases project teams may consist of people from different organisations. Project teams may be inter-disciplinary groups and are likely to lie outer surface the normal administration hierarchies.

The project team will be accountable for delivery of the project end product to some subsidise within or outside the organisation. The full advantage of any project will not become available until the project as been completed.

2.1 Project Classification: In recent lifetime more and more actions have been tackled on a project basis. Project teams and a project management move towards have become common in most organisations. The basic approaches to project management remain the same regardless of the type of development being considered. You may find it useful to look upon as projects in relation to a number of most important classifications:

a) **Engineering and construction** the projects are troubled with produce a clear physical output, such as roads, bridges or buildings. The needs of a project team are well defined in stipulations of skills and background, as are the main actions that have to be undergone. Most of the troubles which may confront the project team are likely to have occurred before and therefore their solution may be based upon history experiences.

b) **Foreword of new systems** these projects would consist of computerisation projects and the introduction of new systems and procedures including financial systems. The nature and instrument of government of a project team may vary with the subject of the scheme, as different skills may be required and different end-users may be involved. Major projects involving a systems analysis approach may slot in clearly defined procedures within an organisation.

c) **Responding to deadlines and change** an example of responding to a time limit is the preparation of an annual report by a particular date. An increasing number of projects are concerned with designing organisational or environmental changes, involving developing new goods and services.

2.2 Project Management Tools and techniques:

Project planning is at the heart of project organization. One can't manage and manage project activities if there is no plan.

Without a graph it is impossible to know if the correct manners are underway, if the available possessions are adequate or of the project can be completed within the desired time. The plan becomes the roadmap that the project team member use in the direction of guide them from side to side the scheme activities. Project organization tools and method assist project managers and their team in carrying out service in all nine knowledge areas. For example, some well-liked time-management tools and method include Gantt charts, project network diagram, and critical path analysis.	**Tools & Techniques**
Integration management	Project selection methods, project organization methodologies, stakeholder analyses, project charters, project organization plans, project organization software, change needs change control boards, project review meeting, lessons-learned information
Scope management	Scope statements, work breakdown structures, brain maps, statements of work, needs analyses, scope management plans, scope confirmation techniques, and scope change wheel
Cost Management	Net their value, homecoming on investment, reimburse analyses, earned value organization, project portfolio management, cost estimate, cost management strategy, cost baselines

Time management	Gantt charts, plan network diagrams, critical-path analyses, crashing, fast track, schedule performance capacity
Human resource management	Motivation techniques, empathic listening, accountability assignment matrix, project organizational charts, resource histograms, players building exercises
Quality management	Quality metrics, checklists, quality be in incriminate of charts, Pareto diagrams, fishbone diagrams, maturity model, statistical system.
	Risk management
	Communication management
	Procurement management
	the a variety of internal events of the contracting party, and to form close links with the listed law-making body, is essential in ensure that the key issues of cost, time, excellence, and above all, client approval, can be realized. In whatever field, a triumphant project manager must be able to imagine the entire project from start to finish and to have the aptitude to ensure that this vision is realized. When they are chosen, project manager should be given terms of orientation that define their: Objectives; Responsibilities; Limits of authority.

3CONCLUSION:-

3.1 Responsibilities of a Project Manager: The object of every project manager is to deliver the product on time, within financial plan and with the required quality. Although the precise responsibilities of a project manager will vary from business to company and from project to project, they should always include planning and forecasting. Three additional areas of management accountability are: interpersonal responsibilities, which comprise: leading the project team; - liaising with initiators, senior administration and suppliers; - person the 'figurehead', i.e. setting the example to the development team and representing the project on formal occasion.

informational responsibilities, which include: monitoring the presentation of staff and the implementation of the project plan; - disseminate information about tasks to the project team; - disseminating information about scheme status to initiators and senior management; - performing as the spokesman for the project team.

decisional errands, which include: allocate resources according to the project plan, and adjust those allocations when circumstances speak (i.e. the project manager has responsibility for the budget); - negotiate with the initiator about the optimum interpretation of contractual obligations, with the corporation management for resources, and with project staff concerning their tasks; - handling disturbances to the smooth development of the project such as equipment failures and personnel problems

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